

Jenkins-Practical

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1. Agent Setup

- Launch one EC2 instance and use it as the Master instance, then install Jenkins on this instance.

Instance summary for i-09f46556e4e6cc56a (Master Instance) Info

Updated less than a minute ago

[Connect](#) [Instance state](#) [Actions](#)

Instance ID i-09f46556e4e6cc56a	Public IPv4 address 51.20.92.167 open address	Private IPv4 addresses 10.0.2.129
IPv6 address -	Instance state Running	Public DNS ec2-51-20-92-167.eu-north-1.compute.amazonaws.com open address
Hostname type IP name: ip-10-0-2-129.eu-north-1.compute.internal	Private IP DNS name (IPv4 only) ip-10-0-2-129.eu-north-1.compute.internal	Elastic IP addresses -
Answer private resource DNS name -	Instance type t3.micro	AWS Compute Optimizer finding Opt-in to AWS Compute Optimizer for recommendations. Learn more
Auto-assigned IP address 51.20.92.167 [Public IP]	VPC ID vpc-024f8931266570790 (MyVPC) open	Auto Scaling Group name -
IAM role -	Subnet ID subnet-02c17cda1d166a85b (Public-Subnet) open	

```
root@ip-10-0-2-129:/home/ubuntu# systemctl status jenkins
● jenkins.service - Jenkins Continuous Integration Server
   Loaded: loaded (/usr/lib/systemd/system/jenkins.service; enabled; preset: enabled)
   Active: active (running) since Sat 2026-02-21 09:42:12 UTC; 55min ago
     Main PID: 4513 (java)
        Tasks: 42 (limit: 1015)
      Memory: 428.7M (peak: 598.4M)
         CPU: 1min 20.780s
    CGroup: /system.slice/jenkins.service
            └─4513 /usr/bin/java -Djava.awt.headless=true -jar /usr/share/java/jenkins.war --webroot=/var/

Feb 21 09:42:10 ip-10-0-2-129 jenkins[4513]: 2026-02-21 09:42:10.135+0000 [id=32] INFO jenkins
Feb 21 09:42:10 ip-10-0-2-129 jenkins[4513]: 2026-02-21 09:42:10.174+0000 [id=30] INFO jenkins
Feb 21 09:42:10 ip-10-0-2-129 jenkins[4513]: 2026-02-21 09:42:10.782+0000 [id=31] INFO h.p.b.
Feb 21 09:42:12 ip-10-0-2-129 jenkins[4513]: 2026-02-21 09:42:12.425+0000 [id=31] INFO jenkins
Feb 21 09:42:12 ip-10-0-2-129 jenkins[4513]: 2026-02-21 09:42:12.426+0000 [id=31] INFO jenkins
Feb 21 09:42:12 ip-10-0-2-129 jenkins[4513]: 2026-02-21 09:42:12.498+0000 [id=32] INFO jenkins
Feb 21 09:42:12 ip-10-0-2-129 jenkins[4513]: 2026-02-21 09:42:12.507+0000 [id=30] INFO jenkins
Feb 21 09:42:12 ip-10-0-2-129 jenkins[4513]: 2026-02-21 09:42:12.552+0000 [id=33] INFO jenkins
Feb 21 09:42:12 ip-10-0-2-129 jenkins[4513]: 2026-02-21 09:42:12.585+0000 [id=24] INFO hudson
Feb 21 09:42:12 ip-10-0-2-129 systemd[1]: Started jenkins.service - Jenkins Continuous Integration Server.
lines 1-20/20 (END)
```

- Launch another EC2 instance to act as the Worker Node and name it as the Agent instance.

Instance summary for i-0b9c8acafb43f4d62 (Agent Instance) [Info](#)

Updated less than a minute ago

[Connect](#) [Instance state](#) [Actions](#)

Instance ID i-0b9c8acafb43f4d62	Public IPv4 address 51.21.2.190 open address	Private IPv4 addresses 10.0.15.217
IPv6 address -	Instance state Running	Public DNS ec2-51-21-2-190.eu-north-1.compute.amazonaws.com open address
Hostname type IP name: ip-10-0-15-217.eu-north-1.compute.internal	Private IP DNS name (IPv4 only) ip-10-0-15-217.eu-north-1.compute.internal	Elastic IP addresses -
Answer private resource DNS name -	Instance type t3.micro	AWS Compute Optimizer finding Opt-in to AWS Compute Optimizer for recommendations. Learn more
Auto-assigned IP address 51.21.2.190 [Public IP]	VPC ID vpc-024f8931266570790 (MyVPC)	Auto Scaling Group name -
IAM role -	Subnet ID subnet-02c17cda1d166a85b (Public-Subnet)	

- To connect the Agent instance to the Master instance, generate an SSH key pair on the Master node using ssh-keygen. This generates a private key (keep it secure) and a public key (to be shared with the Agent).

```
root@ip-10-0-2-129:/home/ubuntu/jenkins-tutorial/3-java-jenkins-docker-app# ssh-keygen
Generating public/private ed25519 key pair.
Enter file in which to save the key (/root/.ssh/id_ed25519):
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /root/.ssh/id_ed25519
Your public key has been saved in /root/.ssh/id_ed25519.pub
The key fingerprint is:
SHA256:1QeGA6D8vac3pq1aYNhgFgFwQ2y6YjKqBhDYayjdu60 root@ip-10-0-2-129
The key's randomart image is:
+--[ED25519 256]--+
|o==   .... .o   |
|o.++ .   oo .   |
| * o+   ... .   |
|= + . = . . .   |
|oo o. = S       |
|* . + . .       |
|* . +o o .       |
|.. o. . . ++     |
|+   E...+ = ..   |
+-----[SHA256]-----+
```

```

root@ip-10-0-2-129:~# cd /root/.ssh
root@ip-10-0-2-129:~/.ssh# ls
authorized_keys  id_ed25519  id_ed25519.pub
root@ip-10-0-2-129:~/.ssh#

```

```

root@ip-10-0-2-129:~/.ssh# ls
authorized_keys  id_ed25519  id_ed25519.pub
root@ip-10-0-2-129:~/.ssh# cat id_ed25519
-----BEGIN OPENSSH PRIVATE KEY-----
b3BlbnNzaC1rZXktdjEAAAABG5vbmUAAAABbm9uZQAAAAAAAAABAAAAMwAAAAAzC2gtZW
QyNTUxOQAAACABT3NP3cVLsH1y/BEiXUap9Q0TU5ePzSvE8oWzD2ey0wAAAAJi0GpCHtBqQ
hwAAAAAzC2gtZWQyNTUxOQAAACABT3NP3cVLsH1y/BEiXUap9Q0TU5ePzSvE8oWzD2ey0w
AAAEtDcNo5p4I0ZJziHftP447U52eOmsbkPB9d1J8iyPWwtwG3c0/dxUuwfXL8ESJdRqn1
DRNTl4/NK8TyhbMPZ7LTAAAAEnJvb3RAaXAtMTAtMCM0yLTEyOQECAw==
-----END OPENSSH PRIVATE KEY-----
root@ip-10-0-2-129:~/.ssh# cat id_ed25519.pub
ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAIAG3c0/dxUuwfXL8ESJdRqn1DRNTl4/NK8TyhbMPZ7LT root@ip-10-0-2-129
root@ip-10-0-2-129:~/.ssh#

```

- On the Agent instance, add the public key to the `authorized_keys` file. Then update the system and install OpenJDK on the Agent instance.

```

2. Master Instance 3. Agent Instance
ubuntu@ip-10-0-15-217:~$ cd ~/.ssh/
ubuntu@ip-10-0-15-217:~/.ssh$ ls
authorized_keys
ubuntu@ip-10-0-15-217:~/.ssh$ cat authorized_keys
ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAQCA7HaiF3cttkIGAhszd7RPwTuaLzzToSZ0v1FiHTffzkT0jyMxBVb93sUb6G5ldUz48T4/0
5ETk7io0Ab4MXkf1BVL/yN0SncRgU07i7F23bP60m0rVrPW1gepKVSASTCuHJFLtJ/mg0nfudQ1ahrePwaZEv0+sw20ME3E93Kb+gxCoKHgs
VksdZvAmg0Cg0lnf0vW/XipNkLFWZKddYv0jCQWLQU5bGwXL5Q0DQ53RH34DaxidLwRXenCvgo4kDM8kGAGygSd0X3Yyt0KecELHd9Ezjwr
COMqVGX0yb6Mi9WnvIbV6t4D6X163SRZQ0XedDh9FY0e8TGf5F/W7vwd linux1
ubuntu@ip-10-0-15-217:~/.ssh$ echo "ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAIAG3c0/dxUuwfXL8ESJdRqn1DRNTl4/NK8Ty
hbMPZ7LT root@ip-10-0-2-129" >> authorized_keys
ubuntu@ip-10-0-15-217:~/.ssh$

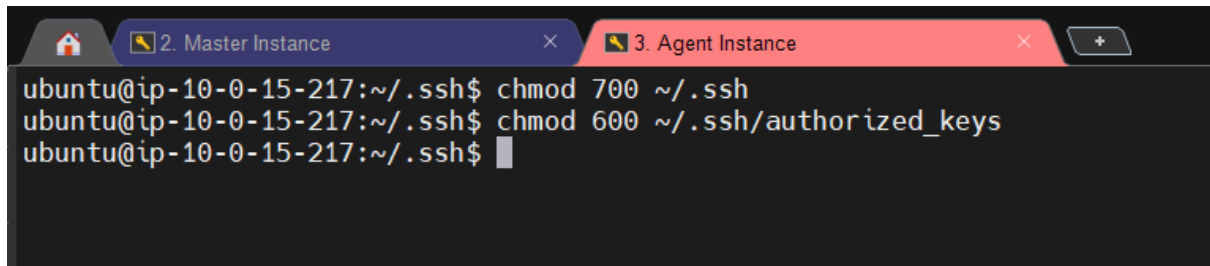
```

```

ubuntu@ip-10-0-15-217:~/.ssh$ java --version
openjdk 17.0.18 2026-01-20
OpenJDK Runtime Environment (build 17.0.18+8-Ubuntu-124.04.1)
OpenJDK 64-Bit Server VM (build 17.0.18+8-Ubuntu-124.04.1, mixed mode, sharing)
ubuntu@ip-10-0-15-217:~/.ssh$

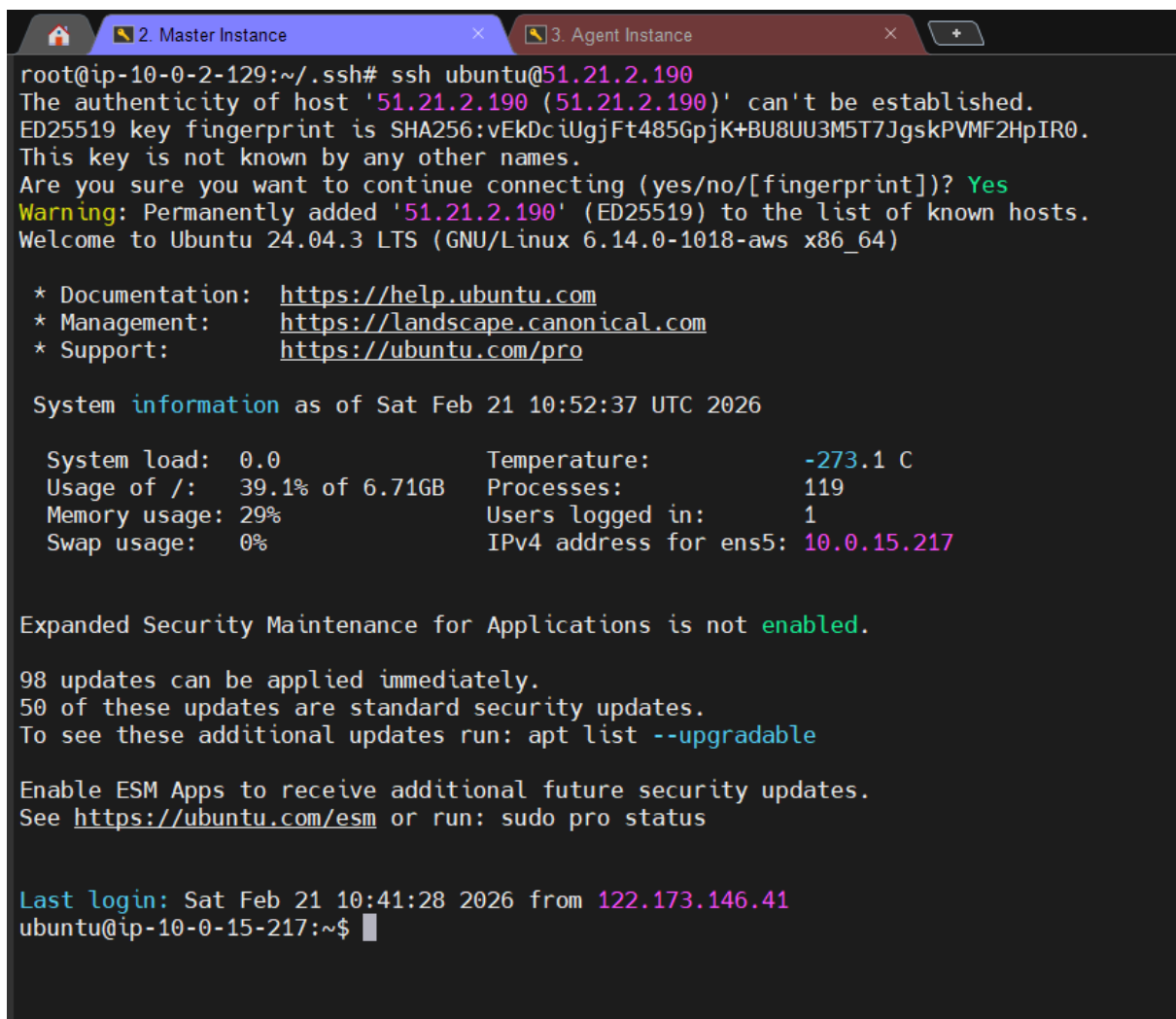
```

- Set the required permissions for the .ssh directory (read, write, and execute) and for the authorized_keys file (read and write) on the Agent node.



```
ubuntu@ip-10-0-15-217:~/.ssh$ chmod 700 ~/.ssh
ubuntu@ip-10-0-15-217:~/.ssh$ chmod 600 ~/.ssh/authorized_keys
ubuntu@ip-10-0-15-217:~/.ssh$
```

- Verify the Agent connection from the Master instance by performing an SSH connection using the public IP address of the Agent instance.



```
root@ip-10-0-2-129:~/.ssh# ssh ubuntu@51.21.2.190
The authenticity of host '51.21.2.190 (51.21.2.190)' can't be established.
ED25519 key fingerprint is SHA256:vEkDciUgjFt485GpjK+BU8UU3M5T7JgskPVMF2HpIR0.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? Yes
Warning: Permanently added '51.21.2.190' (ED25519) to the list of known hosts.
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.14.0-1018-aws x86_64)

* Documentation:  https://help.ubuntu.com
* Management:    https://landscape.canonical.com
* Support:       https://ubuntu.com/pro

System information as of Sat Feb 21 10:52:37 UTC 2026

System load:  0.0           Temperature:    -273.1 C
Usage of /:   39.1% of 6.71GB Processes:      119
Memory usage: 29%          Users logged in: 1
Swap usage:   0%           IPv4 address for ens5: 10.0.15.217

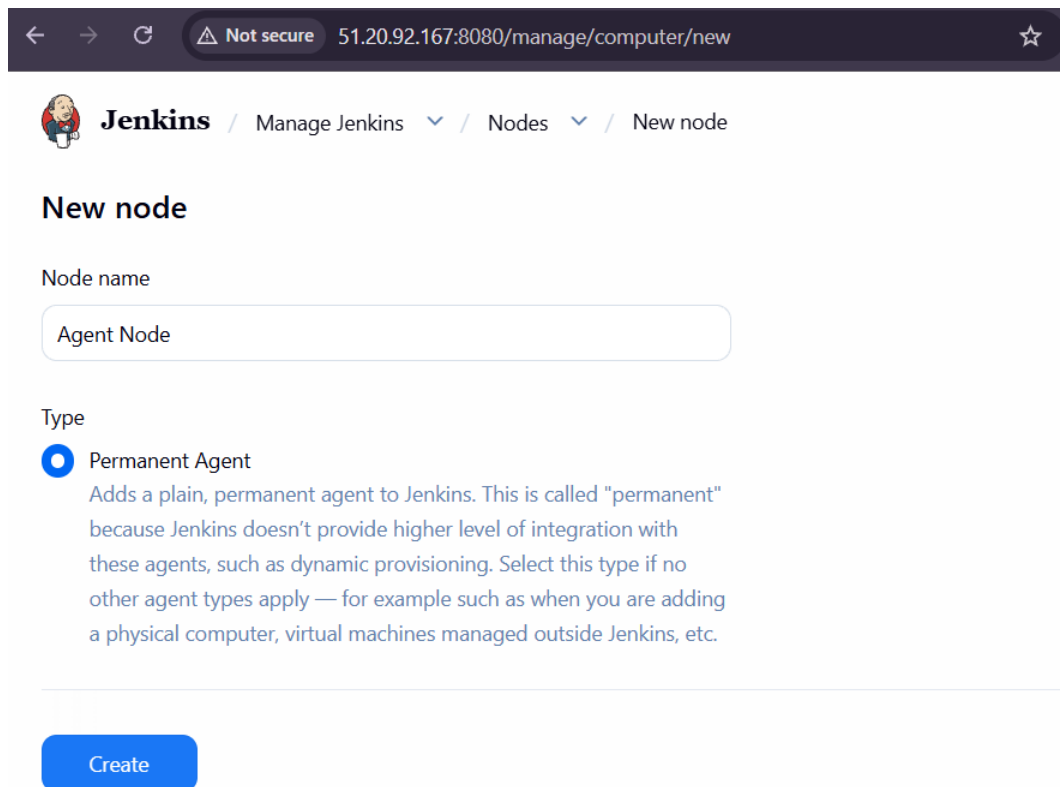
Expanded Security Maintenance for Applications is not enabled.

98 updates can be applied immediately.
50 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

Last login: Sat Feb 21 10:41:28 2026 from 122.173.146.41
ubuntu@ip-10-0-15-217:~$
```

- Log in to the Jenkins web interface from the Master instance and add the Agent node configuration.



← → ↻ ⚠ Not secure 51.20.92.167:8080/manage/computer/new ☆

 **Jenkins** / Manage Jenkins ▾ / Nodes ▾ / New node

New node

Node name

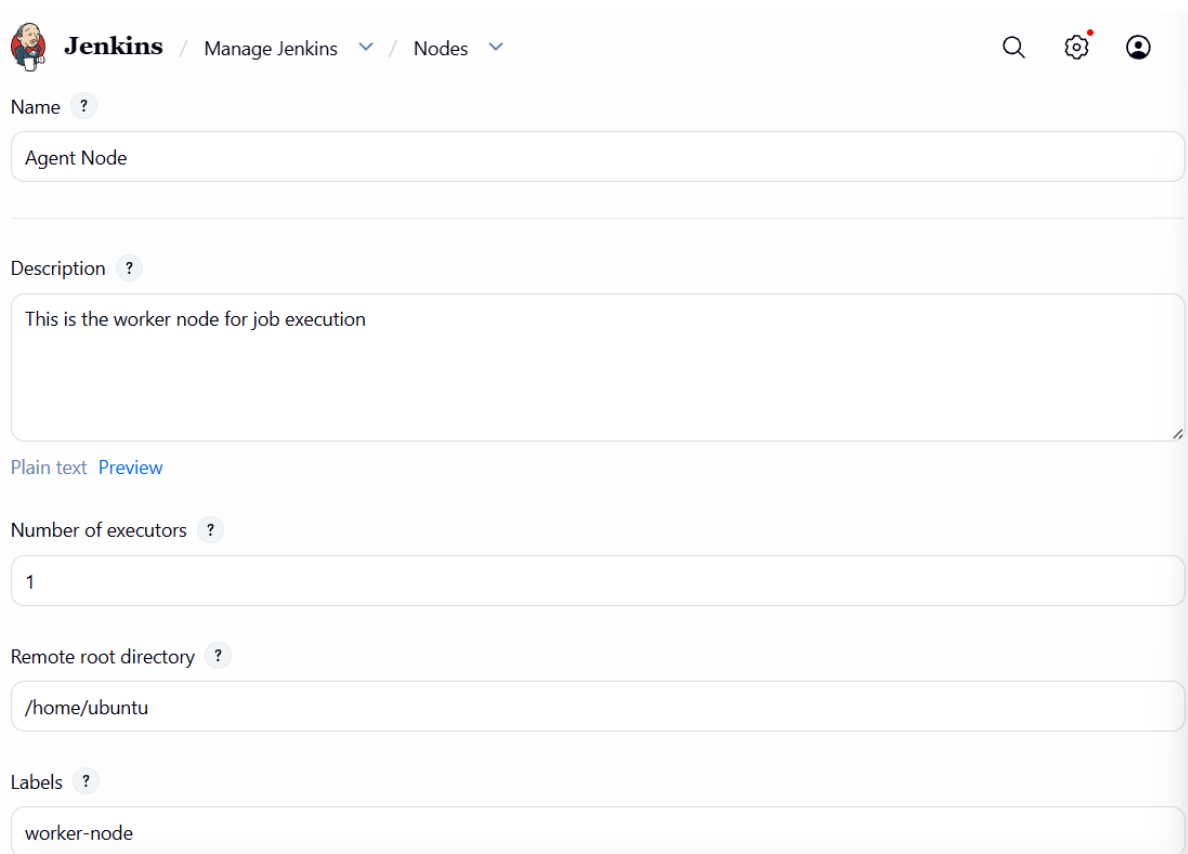
Agent Node

Type

☒ Permanent Agent

Adds a plain, permanent agent to Jenkins. This is called "permanent" because Jenkins doesn't provide higher level of integration with these agents, such as dynamic provisioning. Select this type if no other agent types apply — for example such as when you are adding a physical computer, virtual machines managed outside Jenkins, etc.

Create



 **Jenkins** / Manage Jenkins ▾ / Nodes ▾ 🔍 ⚙️ 👤

Name ?

Agent Node

Description ?

This is the worker node for job execution

Plain text [Preview](#)

Number of executors ?

1

Remote root directory ?

/home/ubuntu

Labels ?

worker-node

Usage ?

Use this node as much as possible

Launch method ?

Launch agents via SSH

Host ?

51.21.2.190

Credentials ?

- none -

+ Add

 The selected credentials cannot be found

< Add SSH Username with private key

×

Scope ?

Global (Jenkins, nodes, items, all child items, etc)

ID ?

worker-node

Description ?

Worker-jenkins-agent

Username

ubuntu

☐ Treat username as secret ?

<

Add SSH Username with private key

x

☐ Treat username as secret ?

Private Key

☒ Enter directly

Key

Enter New Secret Below

-----BEGIN OPENSSH PRIVATE KEY-----
b3B1bnNzaC1rZXktbjEAAAAAG5vbmUAAAAEbm9uZQAAAAAAAAABAAA
AMwAAAAtzc2gtZW
QyNTUxOQAAACABt3NP3cVLsh1y/BEiXUap9Q0TU5ePzSvE8owZD2ey0
wAAAji0GpCHTBqQ
hwAAAAtzc2gtZWQyNTUxOQAAACABt3NP3cVLsh1y/BEiXUap9Q0TU5e
PzSvE8owZD2ey0w
AAAEtNoSp4I0ZJziHftP447U52eOmsbkPB9d1J8iyPWwtwG3c0/dx
UuwfXL8ESJdRqn1
DRNT14/NK8TyhbMPZ7LTAAAEEnJvb3RAaXAtMTAtMC0yLTeyOQECaw=

Passphrase

Create

 **Jenkins**

/ Manage Jenkins

/ Nodes

Launch method ?

Launch agents via SSH

Host ?

51.21.2.190

Credentials ?

ubuntu (Worker-jenkins-agent)

+ Add

Host Key Verification Strategy ?

Non verifying Verification Strategy

Advanced

Availability ?

Keep this agent online as much as possible

Node Properties

Save

← → ↺

Not secure 51.20.92.167:8080/manage/computer/

☆

 Jenkins

Manage Jenkins

Nodes

🔍

⚙️

👤

Nodes

+ New Node

Configure Monitors

↻

Add, remove, control and monitor the various nodes that Jenkins runs jobs on.

S	Name ↓	Architecture	Clock Difference	Free Disk Space	Free Swap Space	Free Temp Space	Response Time	
	Agent Node		N/A	N/A	N/A	N/A	N/A	⚙️
	Built-In Node	Linux (amd64)	In sync	2.62 GiB	0 B	2.62 GiB	0ms	⚙️
Data obtained		4 ms	3 ms	3 ms	3 ms	2 ms	1 ms	

Icon: S M L

Legend

← → ↺

Not secure 51.20.92.167:8080/manage/computer/

☆

 Jenkins

Nodes

Agent Node

Log

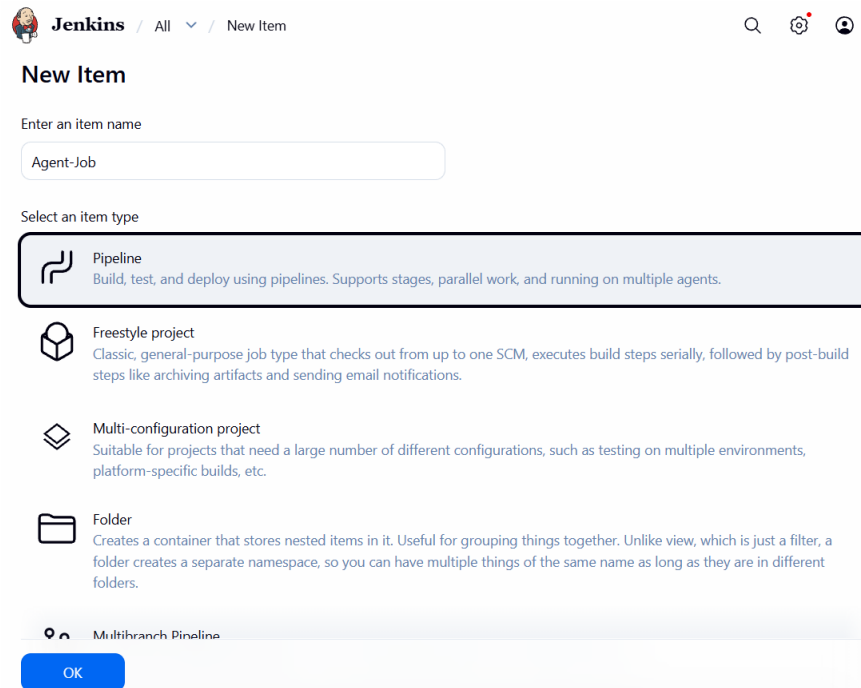
🔍

⚙️

👤

```
_=']'
[02/21/26 11:07:18] [SSH] Starting sftp client.
[02/21/26 11:07:18] [SSH] Copying latest remoting.jar...
[02/21/26 11:07:18] [SSH] Copied 1,407,915 bytes.
Expanded the channel window size to 4MB
[02/21/26 11:07:18] [SSH] Starting agent process: cd
"/home/ubuntu" && java -jar remoting.jar -workDir
/home/ubuntu -jar-cache /home/ubuntu/remoting/jarCache
Feb 21, 2026 11:07:18 AM
org.jenkinsci.remoting.engine.WorkDirManager initializeWorkDir
INFO: Using /home/ubuntu/remoting as a remoting work directory
Feb 21, 2026 11:07:18 AM
org.jenkinsci.remoting.engine.WorkDirManager setupLogging
INFO: Both error and output logs will be printed to
/home/ubuntu/remoting
<===[JENKINS REMOTING CAPACITY]===>channel started
Remoting version: 3352.v17a_fb_4b_2773f
Launcher: SSHLauncher
Communication Protocol: Standard in/out
This is a Unix agent
Agent successfully connected and online
```


- Verify the Agent connection by running a Jenkins pipeline to confirm that the Agent connects successfully.



The screenshot shows the Jenkins 'New Item' page. At the top, there's a breadcrumb 'Jenkins / All / New Item'. Below the title 'New Item', there's a text input field labeled 'Enter an item name' with 'Agent-Job' entered. Underneath, a section 'Select an item type' lists four options: 'Pipeline' (highlighted with a blue border), 'Freestyle project', 'Multi-configuration project', and 'Folder'. Each option has a brief description. At the bottom, there's a 'Multibranch Pipeline' option and an 'OK' button.

Jenkins / All / New Item

New Item

Enter an item name

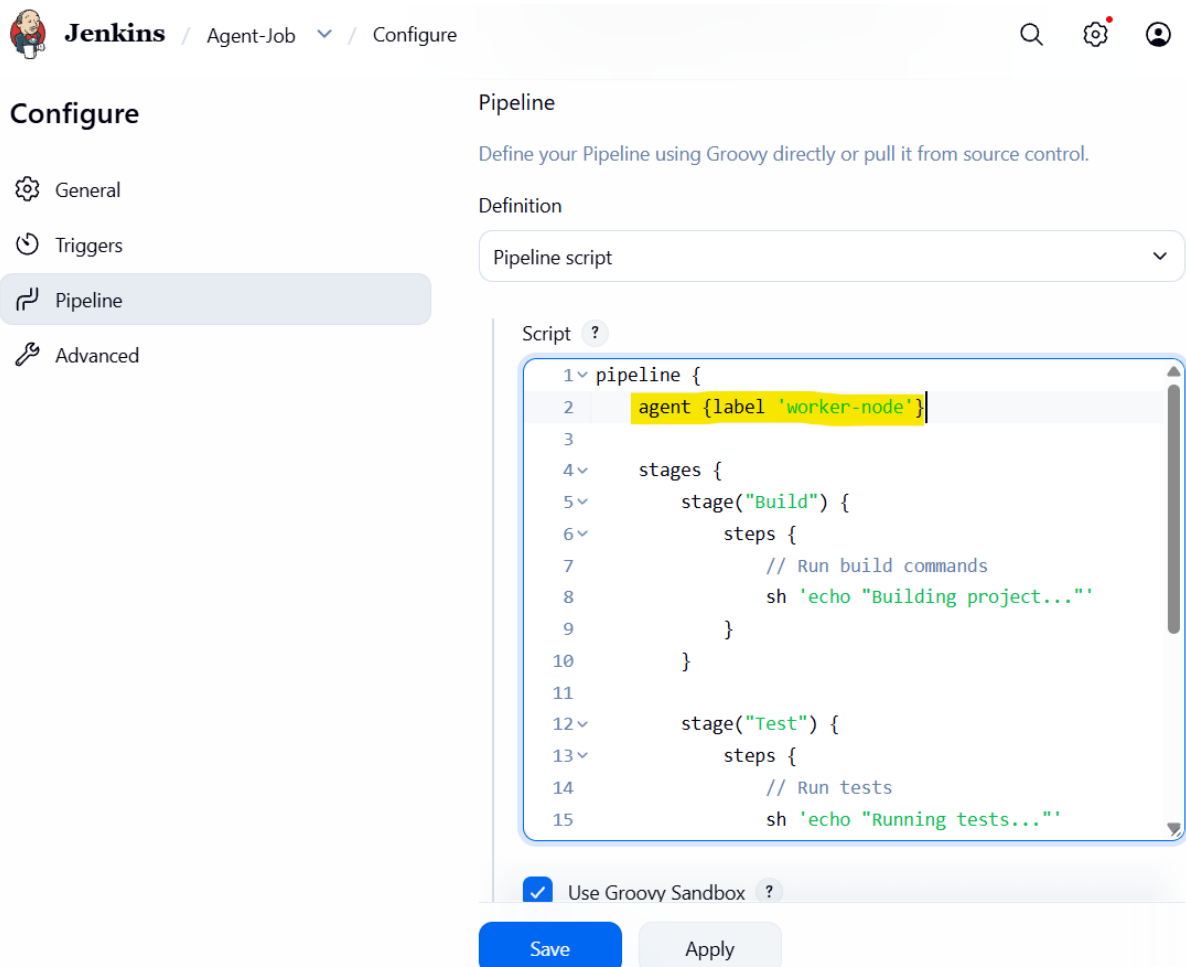
Agent-Job

Select an item type

- Pipeline**
Build, test, and deploy using pipelines. Supports stages, parallel work, and running on multiple agents.
- Freestyle project
Classic, general-purpose job type that checks out from up to one SCM, executes build steps serially, followed by post-build steps like archiving artifacts and sending email notifications.
- Multi-configuration project
Suitable for projects that need a large number of different configurations, such as testing on multiple environments, platform-specific builds, etc.
- Folder
Creates a container that stores nested items in it. Useful for grouping things together. Unlike view, which is just a filter, a folder creates a separate namespace, so you can have multiple things of the same name as long as they are in different folders.

Multibranch Pipeline

OK



The screenshot shows the Jenkins 'Configure' page for the 'Agent-Job'. The left sidebar has 'Configure' at the top, followed by 'General', 'Triggers', 'Pipeline' (selected), and 'Advanced'. The main content area is titled 'Pipeline' and has a subtitle 'Define your Pipeline using Groovy directly or pull it from source control.' Below this, there's a 'Definition' section with a dropdown menu set to 'Pipeline script'. A 'Script' section contains a Groovy pipeline script. At the bottom, there's a checkbox 'Use Groovy Sandbox' which is checked, and two buttons: 'Save' and 'Apply'.

Jenkins / Agent-Job / Configure

Configure

- General
- Triggers
- Pipeline**
- Advanced

Pipeline

Define your Pipeline using Groovy directly or pull it from source control.

Definition

Pipeline script

Script

```
1 pipeline {
2   agent {label 'worker-node'}
3
4   stages {
5     stage("Build") {
6       steps {
7         // Run build commands
8         sh 'echo "Building project..."'
9       }
10    }
11
12    stage("Test") {
13      steps {
14        // Run tests
15        sh 'echo "Running tests..."'
16      }
17    }
18  }
19 }
```

☒ Use Groovy Sandbox

Save Apply



📄 Status

</> Changes

▶ Build Now

⚙️ Configure

🗑 Delete Pipeline

🔍 Full Stage View

📁 Stages

✎ Rename

❓ Pipeline Syntax

Builds >



Today

✅ #2 11:21 AM



Agent-Job

Agent running job

📝 Edit description

Stage View

Average stage times: (full run time: ~2s) #2 Feb 21 16:51 No Changes	Build	Test	Deploy
	890ms	363ms	361ms
	890ms	363ms	361ms

Permalinks